

Ammara Noor

AI / ML Engineer - Computer Vision, LLM & RAG Systems, Production ML

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Available immediately | Open to remote & onsite

SUMMARY

AI/ML engineer who takes systems from data collection through to production deployment. Built and commissioned two systems on a live textile floor: an AI garment size measurement station, and a fabric defect detector at 95% accuracy on a ~1,600-image dataset I collected on-site. Shipped production RAG pipelines into live client applications, and reproduced PatchCore from the original paper to 99.56% AUROC on MVTec AD. 1st place, ICAT National Robotics Competition 2025.

TECHNICAL SKILLS

Computer Vision: Object detection (YOLOv8), image classification, segmentation, anomaly detection (PatchCore), OCR, object tracking, real-time video analytics, image processing, dataset collection and labelling, Roboflow

Machine Learning: PyTorch, TensorFlow, scikit-learn, Hugging Face, Ultralytics, OpenCV, NumPy, Pandas, training, fine-tuning, transfer learning, evaluation (AUROC, mAP, F1, precision/recall), MLflow experiment tracking, threshold calibration

LLM & Generative AI: Retrieval-augmented generation (RAG), embeddings, vector search, chunking and retrieval strategy, prompt engineering, LLM API integration, structured outputs, document processing

MLOps & Deployment: FastAPI, Flask, REST API design, Docker, Linux, Git, ONNX export, latency profiling, inference optimisation, drift monitoring, on-site installation, live camera and hardware integration

Backend, Frontend & Languages: Node.js, Express.js, MySQL, SQLite, role-based authentication, React.js, Next.js, operator dashboards | Python, JavaScript (ES6+), SQL, C/C++ | English (professional working), Urdu (native)

EXPERIENCE

AI / Computer Vision Engineer (Part-Time) - Robionix Technologies, Lahore, Pakistan Feb 2026 - Jun 2026

Built and deployed MagicQC, an AI-based automated garment size measurement system: a fixed-camera station that segments the garment, derives seven points of measure and calls each against the buyer tolerance for one PASS/FAIL verdict. Shipped as a floor desktop app plus a React/Node web app managing brands, article types, operators and purchase orders, and commissioned on the production line at MEB Karachi.

Built a separate fabric defect detection system for the mill floor: YOLOv8 across three known defect classes fused with PatchCore anomaly detection for unseen defects, reaching 95% accuracy on a held-out test set.

Collected and labelled the ~1,600-image defect dataset on-site because none existed; shipped the runtime solo - camera capture, OpenCV preprocessing, dual-model inference, Flask API, React dashboard, Dockerised - then ran operator rollout and re-tuned thresholds under production conditions to cut false positives.

Exhibited MagicQC at the 32nd Textile Asia Expo - Nishat Mills, Gul Ahmed and Sapphire requested evaluations; presented the technology and business case at a national industry open house.

Full-Stack AI Developer - Evolvian Softwares, Lahore, Pakistan Aug 2025 - Jan 2026

Shipped production RAG pipelines end to end - document ingestion, embeddings, vector retrieval, LLM generation - into live client applications, with the REST APIs and React dashboards (role-based auth, live updates) that served them.

Rebuilt chunking and retrieval strategy after diagnosing why standard metrics did not predict real failures, raising output consistency across messy client documents.

Collapsed a multi-step manual review into one automated pipeline by chaining LLM text processing, OCR and YOLOv8 detection; owned features from client requirement to deployed release.

AI & UAV Engineering Intern - National Development Complex (NESCOM), Islamabad, Pakistan Jul 2025 - Sep 2025

Built a UAV surveillance system performing person tracking and vehicle detection on live aerial video feeds.

Delivered a parking operations dashboard turning the raw detection stream into live slot occupancy and searchable footage, filterable by person, vehicle, face or plate.

PROJECTS

Industrial Anomaly Detection - MLOps Pipeline - PyTorch, PatchCore, MLflow, FastAPI, Docker, ONNX github.com/ammara620-de/industrial-anomaly-detection-mlops

Implemented PatchCore from the original paper - frozen WideResNet-50 features, patch memory bank, 1% greedy k-center coreset, trained on defect-free images only. 99.56% AUROC on MVTec AD carpet: 87 of 89 defects caught, 26 of 28 good pieces passed, precision/recall/F1 all 0.9775 at the 95th-percentile threshold.

Served through FastAPI with MLflow-tracked runs, Docker Compose and a drift signal; profiled 535 ms/image on CPU against a 33 ms budget and documented the ONNX/INT8 path.

RallyLens - Sports Analytics - Python, YOLO, OpenCV, object tracking, pose estimation

Player and ball tracking on ordinary handheld coaching footage: identities held through the rally, ball speed derived from the recovered track, shot and rally counts computed rather than tagged by hand. Delivers an annotated video plus a dashboard of per-player counts, ball-speed history and an event timeline.

EDUCATION

BS Computer Engineering - National University of Technology (NUTECH), Islamabad 2022 - 2026

CGPA 3.44 / 4.00 | Coursework: machine learning, deep learning, computer vision, databases, data structures and algorithms, OOP.

ACHIEVEMENTS & LEADERSHIP

1st Place, ICAT National Robotics Competition 2025 | Runner-Up, NAIS 2025 National AI Seminar | Millennium Fellow, United Nations Academic Impact Programme.

Chair, IEEE Women in Engineering, NUTECH Student Branch | Chief Coordinator, NUTECH Student Council | Ambassador, 6th International Student Convention & Expo 2026, Islamabad.